

YAO DONG

Email : dong_yao@hotmail.com Phone : 778-882-7995 **SENIOR SOFTWARE ENGINEER | JAVA BACKEND | DISTRIBUTED SYSTEMS | PERFORMANCE ENGINEERING**

Senior Java Engineer with 15+ years of experience in backend development, enterprise systems integration, and architecture design across commercial and government projects.

CAREER HIGHLIGHTS | KEY ACHIEVEMENTS

Java & JVM Internals

- **JVM Performance & High-Throughput Tuning:** Resolved a critical OutOfMemory (OOM) error by analyzing heap dumps and profiling memory usage under 150K+ Kafka backlog pressure. Eliminated a massive memory spike by implementing backpressure using Java Semaphore and Phaser to limit in-flight threads.
- **Multithreading:** Engineered high-concurrency systems utilizing lock-free algorithms (CAS) and advanced synchronization primitives (AtomicInteger, LongAdder) to eliminate race conditions and maximize throughput under extreme contention.

Distributed Systems & Concurrency

- **Transactional Messaging & Consistency:** Engineered event-driven architectures with Transactional Outbox/Inbox patterns and Kafka's Transactional API (read_committed isolation) to achieve end-to-end **Exactly-Once Semantics (EOS)** across microservice boundaries.
- **Performance Tuning:** Eliminated production **consumer group rebalances** by precisely calibrating `max.poll.interval.ms` and heartbeat intervals under high-throughput conditions; optimized producer throughput by tuning `batch.size` and `linger.ms` to minimize network round-trips.
- **Partitioning & Fault Tolerance:** Designed **business-key-based partition strategies** to guarantee strict per-entity message ordering; also enforced cluster durability via `replication.factor` and `min.insync.replicas` to ensure fault-tolerant data delivery across broker nodes.
- **Distributed Locking & Atomic Operations:** Engineered distributed locking mechanisms using **Redisson (Watchdog)** to solve "lease expiration" concerns; implemented complex business logic via **Lua scripts** to ensure atomicity and minimize RTT overhead.
- **Reliable Messaging & Scheduling:** Architected precision **Delayed Queues** leveraging **Redis ZSET** to complement Kafka's streaming capabilities; integrated Redis as a state provider for **Bucket4j** to implement distributed rate limiting.

Spring | Restful Service | Microservices

- **Resilience & Fault Tolerance:** Architected highly available systems via **Resilience4j** and **Bucket4j**, mastering their underlying algorithms and low-level data structures (e.g., Ring Bucket) to engineer custom, dependency-free fault-tolerance mechanisms from scratch.
- **RESTful API Design:** Standardized cross-service communication using **OpenAPI**. Implemented industry best practices including **Idempotent API** design to safely handle network retries.
- **Distributed Coordination & Security:** Implemented distributed locking via ShedLock/Redis to prevent race conditions in multi-node clusters; secured service-to-service communication utilizing OAuth2/JWT and SSL/TLS termination strategies.

Oracle | Database Performance Tuning | Column-based solution

- **Query Optimization (Mastercard):** Led database performance tuning for a critical claims dispute system, plummeting response times from **400 seconds to 4 seconds**.
- **In-depth Analysis:** Conducted rigorous performance analysis utilizing execution plans, **TKPROF**, partition/indexing strategies, and comprehensive query refactoring.
- **Architectural Enhancement:** Introduced read/write separation based on **CQRS** principles, isolating high-volume query workloads from transactional operations; leveraged Oracle In-Memory Column Store to accelerate analytical workloads on high-volume data sets.

Kubernetes | HELM (AWS/EKS)

- **Zero-Downtime & Resilience:** Engineered deployments by optimizing **pod lifecycles** (readiness probes, preStop hooks, graceful shutdown) to ensure zero-downtime and seamless traffic migration during rolling updates.
- **Resource Optimization:** Resolved **OOMKilled** via `MaxRAMPercentage` and `Request == Limit`; eliminated **CPU Throttling** by omitting CPU limits to preserve JVM burst capacity during GC; applied **low-latency GC** for API pods and **throughput-optimized heap** for Batch pods.

System Architecture & Engineering Principles

- **Distributed System Design:** Applied **CQRS**, **Saga**, and **Inbox/Outbox** patterns to decouple complex domains; navigated **CAP theorem** trade-offs for consistent, horizontally scalable architectures.

- **Scalability & Fault-Tolerant Design:** Designed for **high availability** using stateless services and event-driven communication; applied **leader election** and **distributed locking** for multi-node coordination.

Applied AI Engineering

- **Claude Skills Development:** Built custom Claude Skills, automating operations including remote VM and database command execution.
 - **AI-Augmented Workflows:** Systematically integrated AI-assisted development into daily engineering practice.
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SKILLS

- **Languages:** Java, Python, JavaScript / Node.js
- **Backend Frameworks:** Spring Boot, Spring Cloud, Spring Security, Spring Batch, Resilience4j / Bucket4j
- **Message & Stream Processing:** Apache Kafka, Apache Flink
- **Architecture & Design:** Microservices, Event-Driven Architecture (EDA), CQRS, BPMN / Camunda BPM, RESTful API (OpenAPI / Swagger)
- **Frontend Technologies:** React, Angular
- **Databases & ORM:** Oracle, PostgreSQL, Redis, SQL / NoSQL, Hibernate / iBatis (MyBatis)
- **Performance Tuning:** Oracle SQL Optimization, Execution Plan Analysis, TKPROF
- **Cloud & DevOps:** AWS, Kubernetes (K8s), Jenkins CI/CD, Maven, Git / Bitbucket, Control-M
- **Observability:** Splunk, Dynatrace

- Security & Domain: OAuth, Curam

EXPERIENCE

Mastercard | May 2024 – Now

- Significantly accelerated Oracle query execution times and successfully resolved critical JVM OutOfMemory (OOM) defects in production systems.
- Joined a specialized task force from the larger engineering team to develop machine learning solutions aimed at reducing operational costs.
- Delivered multiple internal demos on practical AI applications in day-to-day engineering, improving team productivity in development and troubleshooting.
- Supported complex dispute workflows by integrating rule-driven decision logic via IBM Operational Decision Manager (ODM).
- Led internal technical sessions to share Oracle SQL tuning principles.

National Bank of Canada | April 2022 – April 2024

- Designed REST APIs with OpenAPI to standardize cross-service communication.
- Implemented Spring Batch with CONTROL-M for sophisticated batch processing.
- Automated database change management using Liquibase and utilized Okta for secure identity management.
- Spearheaded Jenkins/Harness CI/CD pipeline management and created Splunk alerts for proactive error detection.

HSBC | Jan 2020 – April 2022

- Enhanced the customer onboarding process via middle-tier service improvements for online account opening.
- Maintained application robustness through meticulous unit testing and thorough peer code reviews.

Central 1 Credit Union | Apr 2019 – Jan 2020

- Built a Spring Batch project to load daily external transactions (about 10 million records).
- Built project using Kafka and Apache Flink to dispatch and load large volume data.

Boeing Canada | Dec 2017 – Apr 2019

- Built a microservice ecosystem (Spring Boot, Eureka, Hystrix) to aggregate reported defects across all Boeing airplanes.
- Developed Python ETL scripts for external subsystem data consolidation and implemented an Angular frontend leveraging RxJS and functional programming.

TELUS | Nov 2016 – Dec 2017

- Focused on Kafka platform development and automation testing using Robot Framework.

Boeing Canada | Jan 2014 – Nov 2016

- Collaborated with data scientists to construct the Boeing Fuel Optimizer system utilizing complicated algorithms via Spring, JDBC, and RESTful Services.

TELUS | Jul 2010 – Dec 2013

- Designed and implemented unified enterprise reporting systems leveraging Birt Report, iBatis, EJB, Web Services, and Stripes.

Elastic Path | Jul 2008 – Jul 2010

- Developed extensive eCommerce solutions employing Spring, OpenJPA, Eclipse RCP, Lucene, and Solr.

IBM | Apr 2007 – Jul 2008

- Built scalable web applications for the Alberta government utilizing the Curam framework and EJB to manage various benefits.
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EDUCATION

- **Bachelor of Electronic Engineering**, East China Normal University

CERTIFICATIONS

- Sun Certified Java Programmer
- Curam Certified Developer